Lathi 20题练习册 第3批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

奇偶周期延拓与复信号功率

Lathi 1.3-8 | 原书第 141 页

Boulet 第1章：Periodic Signals, p.8；Finite-Energy and Finite-Power Signals, p.21；Even and Odd Signals, p.23

1.3-8 Let $y_1(t) = y_2(t) = t^2$ over $0 \le t \le 1$. Notice, this statement does not require $y_1(t) = y_2(t)$ for all t.

(a) Define $y_1(t)$ as an even, periodic signal with period $T_1 = 2$. Sketch $y_1(t)$ and determine its power.

(b) Design an odd, periodic signal $y_2(t)$ with period $T_2 = 3$ and power equal to unity. Fully describe $y_2(t)$ and sketch the signal over at least one full period. [*Hint:* There are an infinite number of possible solutions to this problem—you need to find only one of them!]

(c) We can create a complex-valued function $y_3(t) = y_1(t) + jy_2(t)$. Determine whether this signal is periodic. If yes, determine the period T_3. If no, justify why the signal is not periodic.

(d) Determine the power of $y_3(t)$ defined in part (c). The power of a complex-valued function $z(t)$ is

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} z(\tau) z^*(\tau)\, d\tau$$

演算区

冲激导数的广义函数性质

Lathi 1.4-12 | 原书第 142 页

Boulet 第1章：Generalized Functions, p.26

1.4-12 Show that

$$\int_{-\infty}^{\infty} \dot{\delta}(t)\phi(t)\,dt = -\dot{\phi}(0)$$

where $\phi(t)$ and $\dot{\phi}(t)$ are continuous at $t = 0$, and $\phi(t) \to 0$ as $t \to \pm\infty$. This integral defines $\dot{\delta}(t)$ as a generalized function. [*Hint:* Use integration by parts.]

演算区

指数正弦信号的复频率

Lathi 1.4-13 | 原书第 142 页

Boulet 第1章：Exponential Signals，p.9；Periodic Complex Exponential and Sinusoidal Signals，p.17

原题首句的指数记号与随后列出的复频率不一致；按其明确写出的两复频率 sigma+j*omega 和 sigma-j*omega 理解，两项指数分别为 exp((sigma+j*omega)t) 与 exp((sigma-j*omega)t)。原题截图未改写。

1.4-13 A sinusoid $e^{\sigma t}\cos \omega t$ can be expressed as a sum of exponentials e^{st} and e^{-st} [Eq. (1.14)] with complex frequencies $s = \sigma + j\omega$ and $s = \sigma - j\omega$. Locate in the complex plane the frequencies of the following sinusoids:

(a) $\cos 3t$

(b) $e^{-3t}\cos 3t$

(c) $e^{2t}\cos 3t$

(d) e^{-2t}

(e) e^{2t}

(f) 5

演算区

冲激核定义系统的基本性质

Lathi 1.7-10 | 原书第 145 页

Boulet 第1章：System Models and Basic Properties, p.34

1.7-10 A continuous-time system is given by

$$y(t) = 0.5 \int_{-\infty}^{\infty} x(\tau)[\delta(t-\tau) - \delta(t+\tau)]\, d\tau$$

Recall that $\delta(t)$ designates the Dirac delta function.

(a) Explain what this system does.
(b) Is the system BIBO-stable? Justify your answer.
(c) Is the system linear? Justify your answer.
(d) Is the system memoryless? Justify your answer.
(e) Is the system causal? Justify your answer.
(f) Is the system time invariant? Justify your answer.

演算区

离散信号的作图与平均功率

Lathi 3.3-1 | 原书第 316 页

Boulet 第1章：Finite-Energy and Finite-Power Signals，p.21；Discrete-Time Impulse and Step Signals，p.25

3.3-1 Sketch, and find the power of, the following signals:

(a) $(1)^n$

(b) $(-1)^n$

(c) $u[n]$

(d) $(-1)^n u[n]$

(e) $\cos\left[\frac{\pi}{3}n + \frac{\pi}{6}\right]$

演算区

因果正弦、余弦与单位阶跃的卷积

Lathi 2.4-15 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-15 Using direct integration, find $\sin t\,u(t) * u(t)$ and $\cos t\,u(t) * u(t)$.

演算区

非因果有理信号与阶跃的卷积

Lathi 2.4-24 | 原书第 228 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-24 Sketch the functions $x(t) = 1/(t^2 + 1)$ and $u(t)$. Now find $x(t) * u(t)$ and sketch the result.

演算区

有限冲激响应系统对反因果输入的卷积

Lathi 3.8-20 | 原书第 323 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

3.8-20 An LTID system with input $x[n]$ and output $y[n]$ has impulse response $h[n] = 2(u[n+2] - u[n-3])$.

(a) Write a constant-coefficient linear difference equation that has the given impulse response. [*Hint:* First express $h[n]$ in terms of delta functions $\delta[n]$.]

(b) Using graphical convolution, determine the zero-state output of this system in response to the anticausal input $x[n] = 2^n u[-n]$. A simplified closed-form solution is required.

演算区

离散LTI系统的串并联与零状态响应

Lathi 3.8-21 | 原书第 323 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

3.8-21 Consider three LTID systems: system 1 has impulse response $h_1[n] = [\underset{\downarrow}{2}, -3, 4]$, system 2 has impulse response $h_2[n] = [\underset{\downarrow}{0}, 0, -6, -9, 3]$, and system 3 is an identity system (output equals input).

(a) Determine the overall impulse response $h[n]$ if system 1 is connected in cascade with a parallel connection of systems 2 and 3.

(b) For input $x[n] = u[-n]$, determine the zero-state response $y_{zsr}[n]$ of system 2.

演算区

有限序列卷积与滑动法

Lathi 3.8-25 | 原书第 323 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum， p.54

3.8-25 Letting $\downarrow$ identify the $n = 0$ values, use the sliding-tape method to determine the following:

(a) $y_a = [\overset{\downarrow}{2}, 3, -2, -3] * [-10, \overset{\downarrow}{0}, -5]$

(b) $y_b = [2, \overset{\downarrow}{-1}, 3, -2] * [-1, -4, 1, \overset{\downarrow}{-2}]$

(c) $y_c = [\overset{\downarrow}{0}, 0, 3, 2, 1, 2, 3] * [2, 3, -2, \overset{\downarrow}{1}]$

(d) $y_d = [5, 0, 0, \overset{\downarrow}{-2}, 8] * [-1, 1, \overset{\downarrow}{3}, 3, -2, 3]$

(e) $y_e = ([1, \overset{\downarrow}{-1}] * [\overset{\downarrow}{1}, -1]) * ([\overset{\downarrow}{1}, -1] * [1, \overset{\downarrow}{-1}])$

(f) $y_f = ([2, \overset{\downarrow}{-1}] * [\overset{\downarrow}{1}, -2]) * ([\overset{\downarrow}{1}, -2] * [2, \overset{\downarrow}{-1}])$

Outside the values shown, assume all signals are zero.

演算区

从零输入响应判断可能的特征方程

Lathi 2.2-11 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

2.2-11 A system is described by a constant-coefficient linear differential equation and has zero-input response given by $y_0(t) = 2e^{-t} + 3$.

(a) Is it possible for the system's characteristic equation to be $\lambda + 1 = 0$? Justify your answer.

(b) Is it possible for the system's characteristic equation to be $\sqrt{3}(\lambda^2 + \lambda) = 0$? Justify your answer.

(c) Is it possible for the system's characteristic equation to be $\lambda(\lambda + 1)^2 = 0$? Justify your answer.

演算区

从积分方程建立微分系统并求冲激响应

Lathi 2.3-7 | 原书第 225 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Impulse Response of a Differential LTI System，p.101

2.3-7 A causal LTIC system with input $x(t)$ and output $y(t)$ is described by the constant coefficient integral equation

$$y(t) + \int 3y(t)\,dt + \iint 2y(t)\,dt = \iint x(t)\,dt - \iiint x(t)\,dt.$$

(a) Express this system as a constant coefficient linear differential equation in standard operator form.

(b) Determine the characteristic modes of this system.

(c) Determine the impulse response $h(t)$ of this system.

演算区

离散系统零输入响应的闭式表达

Lathi 3.6-1 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

3.6-1 Given $y_0[-1] = 3$ and $y_0[-2] = -1$, determine the closed-form expression of the zero-input response $y_0[n]$ of an LTID system described by the equation $y[n] + \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = \frac{1}{3}x[n] + \frac{2}{3}x[n-2]$.

演算区

Fibonacci数列的差分模型、特征根与稳定性

Lathi 3.6-6 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96；Characteristic Polynomials and Stability， p.112

3.6-6 Leonardo Pisano Fibonacci, a famous thirteenth-century mathematician, generated the sequence of integers

$$\{0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots\}$$

while addressing, oddly enough, a problem involving rabbit reproduction. An element of the Fibonacci sequence is the sum of the previous two.

(a) Find the constant-coefficient difference equation whose zero-input response $f[n]$ with auxiliary conditions $f[1] = 0$ and $f[2] = 1$ is a Fibonacci sequence. Given $f[n]$ is the system output, what is the system input?

(b) What are the characteristic roots of this system? Is the system stable?

(c) Designating 0 and 1 as the first and second Fibonacci numbers, determine the fiftieth Fibonacci number. Determine the one thousandth Fibonacci number.

演算区

连续与离散系统稳定区域的对照设计

Lathi 3.6-9 | 原书第 320 页

Boulet 第3章：Characteristic Polynomials and Stability of Differential and Difference Systems, p.112

3.6-9 Provide a standard-form polynomial $Q(X)$ such that $Q(E)\{y[n]\} = x[n]$ corresponds to a marginally stable third-order LTID system and $Q(D)\{y(t)\} = x(t)$ corresponds to a stable third-order LTIC system.

演算区

周期性、基周期与谐波组成

Lathi 6.1-8 | 原书第 672 页

Boulet 第4章：Linear Combinations of Harmonically Related Complex Exponentials, p.132

6.1-8 State with reasons whether the following signals are periodic or aperiodic. For periodic signals, find the period and state which harmonics are present in the series.

(a) $3\sin t + 2\sin 3t$

(b) $2 + 5\sin 4t + 4\cos 7t$

(c) $2\sin 3t + 7\cos \pi t$

(d) $7\cos \pi t + 5\sin 2\pi t$

(e) $3\cos \sqrt{2}t + 5\cos 2t$

(f) $\sin \frac{5t}{2} + 3\cos \frac{6t}{5} + 3\sin\left(\frac{t}{7} + 30^\circ\right)$

(g) $\sin 3t + \cos \frac{15}{4}t$

(h) $(3\sin 2t + \sin 5t)^2$

(i) $(5\sin 2t)^3$

演算区

反转、平移、调制与线性组合的级数系数

Lathi 6.3-14 | 原书第 675 页

Boulet 第4章：Properties of Continuous-Time Fourier Series，p.139

原题先记输入频谱为 D1[n]，后文又写 X1[n]；两者均指同一输入 x1(t) 的傅里叶级数系数。原题截图保留原记号，作答时统一记号即可。

6.3-14 (a) A 2 rad/s periodic signal $x_1(t)$ has Fourier series spectrum $D_1[n]$. Determine the Fourier series spectrum $X_2[n]$ of $x_2(t) = \frac{1}{3}x_1(-t-5)$ in terms of $X_1[n]$.

(b) A 2 rad/s periodic signal $x_1(t)$ has Fourier series spectrum $D_1[n]$. Determine the Fourier series spectrum $X_2[n]$ of $x_2(t) = \cos(10t)x_1(t)$ in terms of $X_1[n]$.

(c) A 3 rad/s periodic signal $x_1(t)$ has Fourier series spectrum $D_1[n]$. Determine the Fourier series spectrum $X_2[n]$ of $x_2(t) = x_1(-t) - 3x_1(t+2)$ in terms of $X_1[n]$.

演算区

分段二次周期信号的傅里叶级数与幅度谱

Lathi 6.3-15 | 原书第 675 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137；Properties of Continuous-Time Fourier Series，p.139

原文对例6.11的引用是方法提示：使用傅里叶级数性质求非直流系数，不能直接积分；本题的信号条件已全部给出，不依赖该例题的数据。

6.3-15 A 2-periodic signal $x(t)$ is defined as

$$x(t) = \begin{cases} -t^2 - t + 0.25 & -1 \le t < 0 \\ t^2 - t + 0.25 & 0 \le t < 1 \\ x(t+2) & \forall t \end{cases}$$

(a) Plot $x(t)$ over $-2 \le t \le 2$.
(b) Determine D_0, the dc content of $x(t)$.
(c) Similar to Ex. 6.11, use properties and not integration to determine D_n for $n \neq 0$.
(d) Plot the magnitude spectrum $|D_n|$ over a suitable range of n.
(e) How does the magnitude spectrum $|D_n|$ compare to that of signal $y(t) = \cos(\pi t)$? Note similarities as well as major differences.

演算区

正弦乘积的级数频谱与理想滤波

Lathi 6.4-4 | 原书第 675、676 页

Boulet 第4章：Properties of Continuous-Time Fourier Series， p.139；Steady-State Response of an LTI System to a Periodic Signal， p.155

本题跨两页；(c)小题所需原图 P6.4-4 已完整附上。

6.4-4 (a) Find the exponential Fourier series for a signal $x(t) = \cos 5t \sin 3t$. You can do this without evaluating any integrals.

(b) Sketch the Fourier spectra.

(c) The signal $x(t)$ is applied at the input of an LTIC system with frequency response, as shown in Fig. P6.4-4. Find the output $y(t)$.

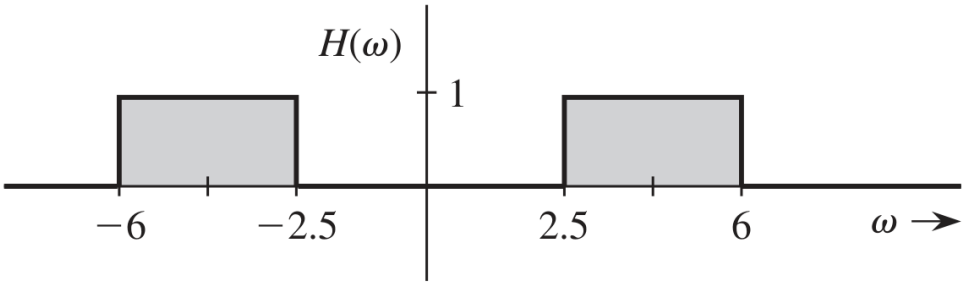


Figure P6.4-4

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信号的最优投影、误差能量与正交性

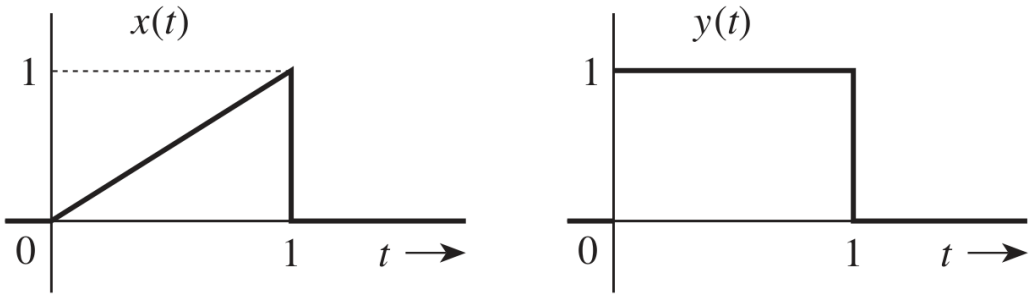
Lathi 6.5-4 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

本题所需原图 P6.5-4(a)、(b)均已附上。

6.5-4 (a) For the signals $x(t)$ and $y(t)$ depicted in Fig. P6.5-4, find the component of the form $y(t)$ contained in $x(t)$. In other words, find the optimum value of c in the approximation $x(t) \approx cy(t)$ so that the error signal energy is minimum.

(b) Find the error signal $e(t)$ and its energy E_e. Show that the error signal is orthogonal to $y(t)$, and that $E_x = c^2 E_y + E_e$. Explain this result in terms of vectors.



(a) (b)

Figure P6.5-4

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